Which of the following statements about AlexNet (Krizhevsky et al., 2012) are correct?

Select one or more:

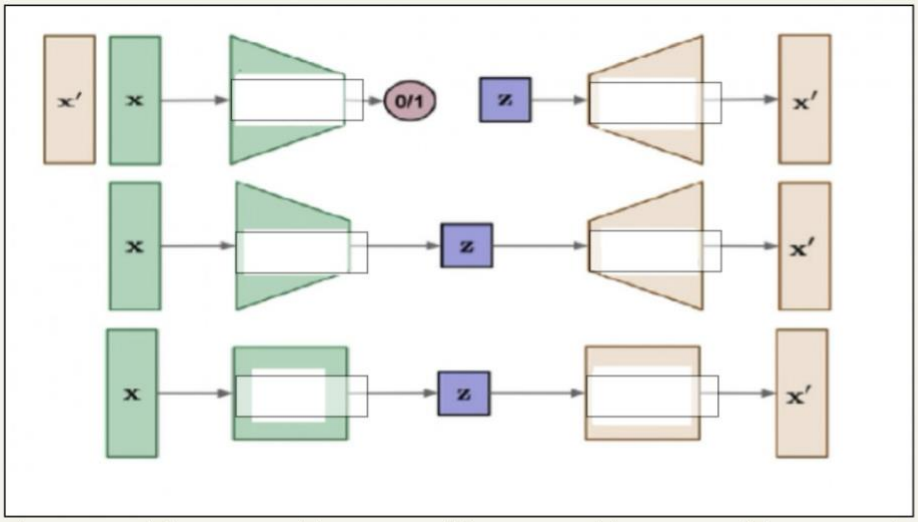
- ☐ a. AlexNet does not have fully-connected layers.
- ☐ b. AlexNet uses strided convolutions instead of max-pooling for down-sampling.
- ☐ c. AlexNet is a deep convolutional network with more than 1 hidden layer.
- ☐ d. One of the most important components of AlexNet is BatchNorm.
- ☐ e. AlexNet uses input images with a resolution of 224x224 pixels and three channels.

Mark the correct statments related to the Universal Approximator Theorem (UAT) for Neural Networks (NNs).

Select one or more:

- ☐ a. In the case of a real-valued, continuous function h defined on $[-\pi, \pi]^2$, the existence of an approximating NN can be derived with the help of Fourier series.
- ☐ b. The UAT is a theoretical statement about the expressive power of NNs.
- ☐ c. The UAT gives a useful algorithm to compute the optimal weights of a NN that should approximate a given continuous function.
- ☐ d. The algorithm related to the UAT is easy to understand and implement, it is however very costly and therefore it is hardly used in practise.
- ☐ e. The UAT shows that any function can be approximated by any given NN with arbitrary precision, if the weights are chosen appropriately. This means that there are no restriction on the function to be approximated and on the structure of the network that should do the job.

In this picture you can see three categories of generative models we discussed in the lecture. Annotate the boxes in the figure with the following terms, so that the main building blocks/networks are assigned correctly: "Generator", "Discriminator", "Encoder", "Decoder", "Flow", "Inverse Flow". Move the boxes on the bottom in the correct way into the grey boxes in the main figure. To be clear, a rectangular shape of the network means that input and output dimensions stay the same, whereas a trapezoidal shape means decreasing (or increasing) dimension in the respective directions.



Generator	Discriminator	Encoder	Decoder	Flow	Inverse Flow

Which of the following statements about autoencoders (**not** including variational autoencoders) are correct?

Select one or more:

- ☐ a. The Input-Output-Jacobian of an autoencoder is always a lower-triangular matrix.
- ☐ b. Autoencoders do not map a single input data point to a single point in latent space, but they map a single input data point to a distribution in latent space.
- ☐ c. Autoencoders are always fully-connected networks.
- ☐ d. All autoencoders use fewer neurons in the hidden layers than in the input layer.
- ☐ e. Autoencoders consist of an encoder and a decoder, which are parametrized neural networks.

Using the notation introduced in the lecture, which of the following objective functions are correct formulations of the standard objective function of Generative Adversarial Networks (GANs)?

Select one:

- a. $$\min_G \max_D \square(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D(\mathbf{z}))]$$

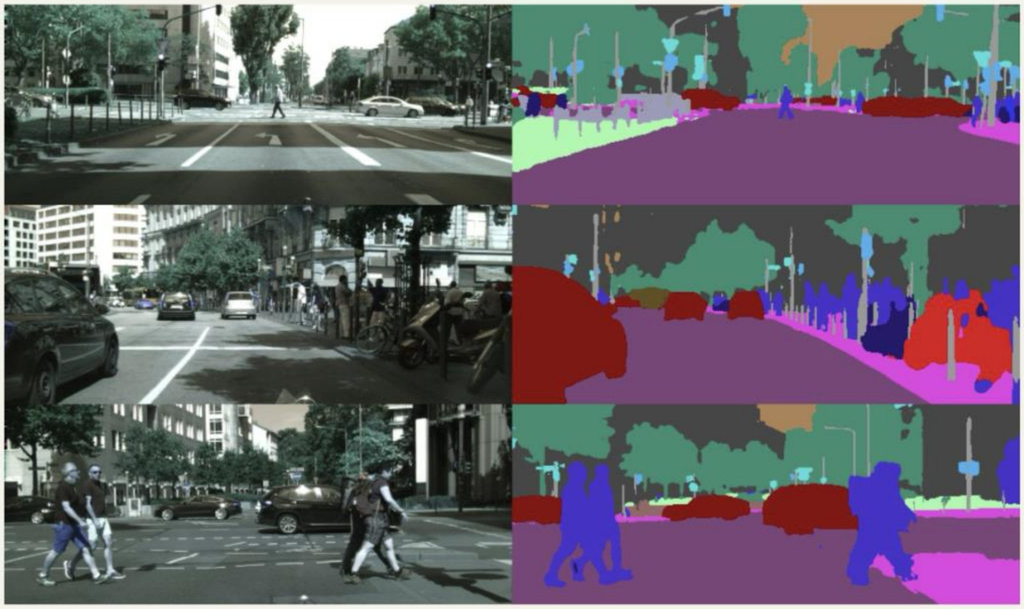
- b. $$\min_G \max_D \square(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{x} \sim p_r} [\log(1 - D(G(\mathbf{x})))]$$

- c. $$\min_G \max_D \square(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log G(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - G(D(\mathbf{z})))]$$

- d. $$\min_G \max_D \square(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(1 - D(G(\mathbf{z})))]$$

- e. $$\min_G \max_D \square(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [D(G(\mathbf{z}))]$$

- f. $$\min_G \max_D \square(D, G) = \mathbb{E}_{\mathbf{x} \sim p_r} [\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z} \sim p_z} [\log(D(G(\mathbf{z})))]$$



Imagine you are working on a machine learning project related to self-driving cars. A camera inside a car provides images of road scenes such as the ones in the picture above (left column). Your task is to train a neural network that supplies images as shown on the right hand side of the picture (right column), in which cars, roads, pedestrians, etc are annotated.

Which of the following statements are true in this context?

Select one or more:

- ☐ a. In order to train such a network, labels for each pixel of the training set images are required.
- ☐ b. This represents an object detection task in computer vision.
- ☐ c. Generative adversarial networks are the most appropriate type of networks to approach this task.
- ☐ d. Deep Neural networks cannot be used for this task because there is no appropriate loss function.
- ☐ e. The intersection over union (IoU) would not be an appropriate metric to assess the performance of a model that is trained on this task.

Which of the following statements about Residual Networks (ResNets) are correct?

Select one or more:

- ☐ a. Residual networks use skip-connections which add the representation $\mathbf{h}^l$ of a former layer l to a later layer $l+1$, that is:

$$\mathbf{h}^{l+1} = \mathbf{h}^l + F(\mathbf{h}^l),$$

  where F can be any transformation in a neural network with appropriate dimensions, such as a fully-connected layer.
- ☐ b. Because of skip-connections, very deep Residual Networks can be trained even without normalization techniques.
- ☐ c. Pre-trained Residual Networks cannot be used for transfer learning because the learned features are usually overly specified to the task on which they were trained.
- ☐ d. Residual networks are always convolutional networks and cannot be used in networks that have only fully-connected layers.
- ☐ e. Leaky ReLUs cannot be used in Residual Networks.

Which of the following statements about Deep Learning based **object detection methods** are correct?

Select one or more:

- ☐ a. Faster R-CNN uses a region proposal network rather than the selective search algorithm that was used in R-CNN.
- ☐ b. The operations ROI-pooling and ROI align operate on feature maps and not on the input image.
- ☐ c. R-CNN uses AlexNet to extract features for Regions Of Interest.
- ☐ d. Object detection can be formulated as regression task in a multi-task setting.
- ☐ e. Fast R-CNN has two output layers (also named heads): one for regression and one for classification.

Assume you are given a convolutional neural network with a fixed architecture. You are only allowed to change the parameters of last convolutional layer and you want to strongly increase the receptive field (RF) of each neuron in the last layer.

For each of the following four possible changes "increasing kernel size", "increase number of kernels", "increase stride", or "switch to dilated convolutions" decide whether they strongly increase, slightly increase or do not increase the receptive field (RF) of your network!

Increase stride of convolution	Choose...
Switch to dilated convolutions	Choose...
increase the number of kernels	Choose...
increase kernel size	Choose...

Choose...
no increase of RF
small increase of RF
✓ strong increase of RF

Which of the following statements about variational autoencoders (VAEs) are true?

Select one or more:

- ☐ a. Variational autoencoders have an encoder-decoder structure, where the encoder and decoder must be symmetric.
- ☐ b. The prior distribution of the variables $\mathbf{z}$ in the latent space must be a discrete distribution for VAEs.
- ☐ c. Variational autoencoders maximize a lower bound on the likelihood of the data.
- ☐ d. Variational autoencoders automatically provide the marginal likelihood of a data point $p(\mathbf{x})$ at low computational costs.
- ☐ e. Variational autoencoders cannot be used to generate images.

Suppose that you have a fixed, multi-dimentional, non-Gaussian distribution $p(\boldsymbol{x})$ that you want to approximate with a Gaussian distribution. You decide that you want to approximate this distribution with a Gaussian $q(\boldsymbol{x}) = \square(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ with some mean vector $\boldsymbol{\mu}$ and some covariance matrix $\boldsymbol{\Sigma}$. You now decide to minimize the Kullback-Leibler divergence between the two distributions $\mathrm{KL}(p \,||\, q)$ with respect to $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$.

Which expressions for $\boldsymbol{\mu}$ and $\boldsymbol{\Sigma}$ mimimize the KL-divergence between the two distributions?

(The usual notation of the lecture is used: $\mathbb{E}$ is expectation, $\mathbb{Cov}$ is the covariance, $\boldsymbol{I}$ is the identity matrix)

- ☐ a. $\boldsymbol{\mu} = \mathbb{E}(\boldsymbol{x})$ and $\boldsymbol{\Sigma} = \mathbb{Cov}(\boldsymbol{x})$
- ☐ b. $\boldsymbol{\mu} = \mathbb{E}(\boldsymbol{x})$ and $\boldsymbol{\Sigma} = \boldsymbol{I}$
- ☐ c. $\boldsymbol{\mu} = \mathbb{E}(\boldsymbol{x}^T\boldsymbol{x})$ and $\boldsymbol{\Sigma} = \mathbb{E}(\boldsymbol{x}^T)$
- ☐ d. $\boldsymbol{\mu} = \mathbb{E}(\boldsymbol{x}^T\boldsymbol{x})$ and $\boldsymbol{\Sigma} = \boldsymbol{I}$
- ☐ e. $\boldsymbol{\mu} = \mathbb{E}(\boldsymbol{x})$ and $\boldsymbol{\Sigma} = \mathbb{E}(\boldsymbol{x}\boldsymbol{x}^T)$

Tick the correct statements related to the Inception Score (IS) and the Frechet Inception Distance (FID).

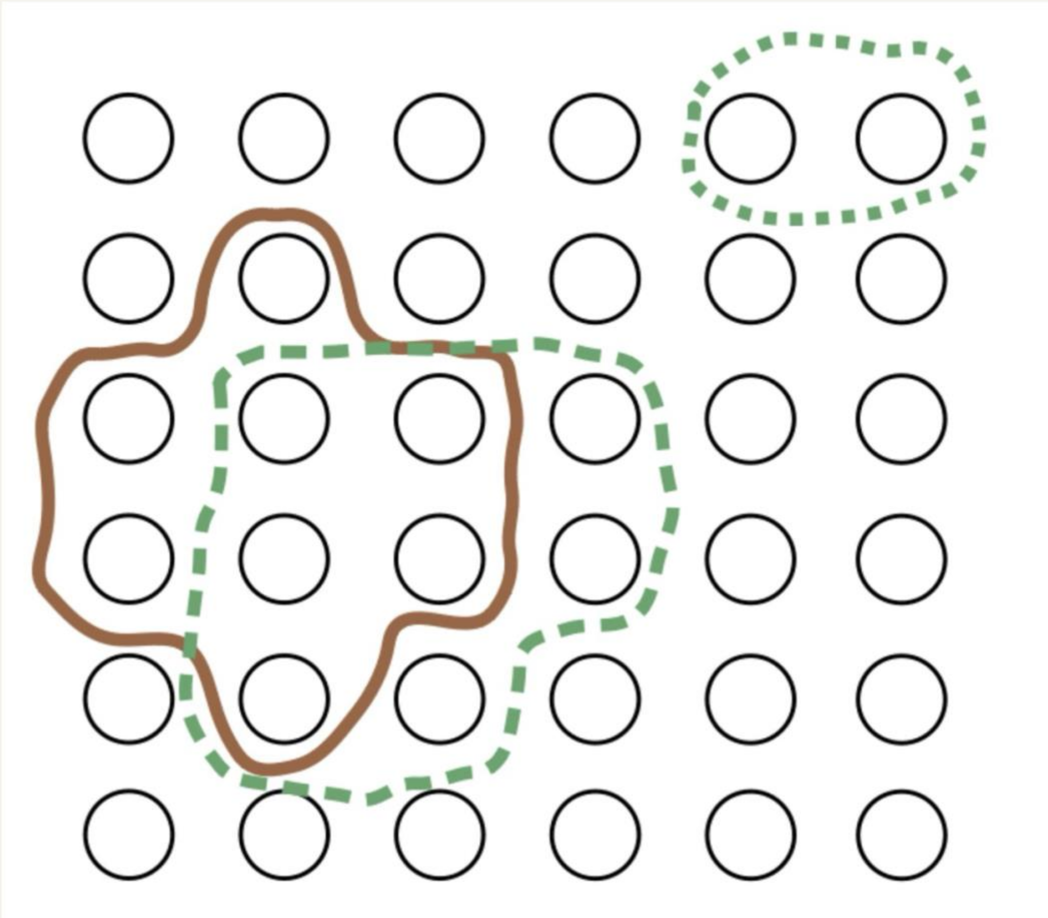
Select one or more:

- [ ] a. The FID can detect mode collapse because mode collapse would influence the covariance matrix of the generated data.
- [ ] b. IS and FID are metrics that assess the quality of GAN-generated images.
- [ ] c. Both IS and FID rely on features for input images that are produced by AlexNet.
- [ ] d. IS and FID can be used as GAN objectives that lead to a faster training procedure.
- [ ] e. The computation of the FID is based on an explicit formula for the Wasserstein 2-distance between two Gauss distributions.

Which of the following statements about normalizing flows are correct?

Select one or more:

- [ ] a. It is impossible integrate multi-layer perceptron as transformation in a normalizing flow because they are not invertible.
- [ ] b. All multi-layer perceptrons with the same input and output dimension can be considered a normalizing flow.
- [ ] c. Normalizing flows cannot contain funtions with a lower triangular Jacobi matrix because they are not invertible.
- [ ] d. Some normalizing flows use functions with a lower triangular Jacobi matrix to facilitate the calculation of the determinant.



In the figure above you see a ground-truth pixels of an object (surrounded by a blue, solid line) and the predicted pixels for this object (surrounded by a green, dotted line). The black circles, that are arranged in a grid, indicate pixels in an image. Calculate the IoU metric for those rectangles! Enter the correct number **rounded to three digits after the comma** into the field below.

Note: If you are in the english version of Moodle, the comma-separator is a dot ".", whereas in the german version the comma-separator is a comma ",".

Answer:

Using the notation introduced in the lecture, which of the following functions are correct formulations of the standard Variational Autoencoder (VAE) objective function?

Select one:

- a. $\square(D,G) = \mathbb{E}_{\mathbf{x}\sim p_r}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[\log(1 - D(G(\mathbf{z})))]$

- b. $\square_{\theta,\phi}(x) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x}|\mathbf{z}) - D_{KL}(p_\phi(\mathbf{x})||p_\theta(\mathbf{z}))$

- c. $\square(D,G) = \mathbb{E}_{\mathbf{x}\sim p_r}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z}[D(G(\mathbf{z})]$

- d. $\square_{\theta,\phi}(x) = \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} \log p_\theta(\mathbf{x}|\mathbf{z}) - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}))$

- e. $\square_\theta(\mathbf{x}) = D_{KL}(p_\mathbf{x}(\mathbf{x};\theta)\|p^*_\mathbf{x}(\mathbf{x}))$

Assume that you are given a two-dimensional vector $\mathbf{u} \sim p_\mathbf{u}(\mathbf{u})$. Moreover, let $T = \begin{pmatrix} 2 & 0 \\ 0 & \frac{1}{2} \end{pmatrix}$ and $\mathbf{x} = T \cdot \mathbf{u}$.

Mark the correct formula for the distribution $p_\mathbf{x}(\mathbf{x})$.

Select one:

- a. $p_\mathbf{u}\left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix} \cdot \mathbf{x}\right) \cdot \frac{1}{2}$

- b. $p_\mathbf{u}\left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 2 \end{pmatrix} \cdot \mathbf{x}\right)$

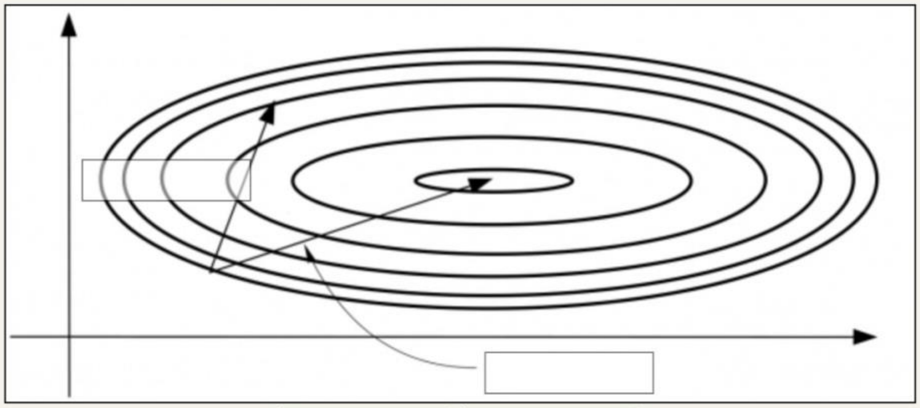
- c. $p_\mathbf{u}\left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} \cdot \mathbf{x}\right) \cdot \frac{1}{2}$

- d. $p_\mathbf{u}\left(\begin{pmatrix} \frac{1}{2} & 0 \\ 0 & 1 \end{pmatrix} \cdot \mathbf{x}\right)$

- e. $p_\mathbf{u}\left(\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \cdot \mathbf{x}\right) \cdot 2$

- f. $p_\mathbf{u}\left(\begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix} \cdot \mathbf{x}\right) \cdot \frac{1}{2}$

In this picture you can see the update direction for Newton's method and the gradient direction for a quadratic error surface (i.e. the level curves origin from a quadratic form like an ellipsis). Annotate the corresponding directions/arrows in the figure with the following terms: “Gradient”, “Newton's Method”. Move the boxes on the bottom in the correct way into the grey boxes in the main figure.



Gradient | Newton's Method

You are given an input $\mathbf{x} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$ and a kernel weight matrix $\mathbf{w} = \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix}$. What is the result of the **transpose convolution operation** $*^T$ (stride 1) of

$$\mathbf{w} *^T \mathbf{x} = ?$$

Select one:

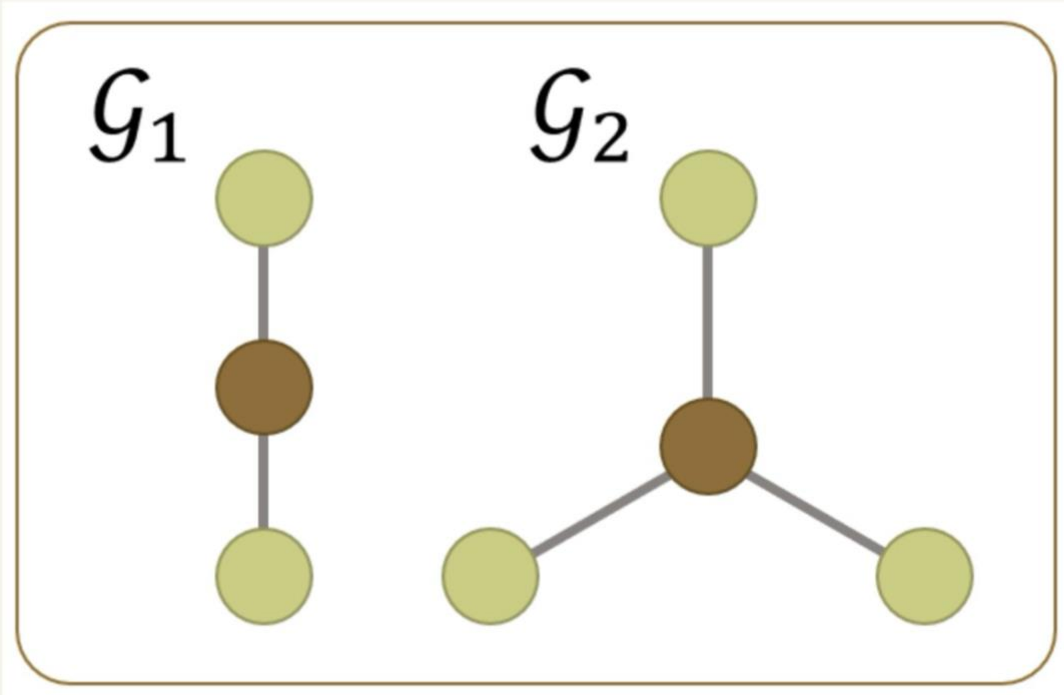
- a. $= \begin{pmatrix} 42 \end{pmatrix}$
- b. $= \begin{pmatrix} 14 \end{pmatrix}$
- c. $= \begin{pmatrix} 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 \end{pmatrix}$
- d. $= \begin{pmatrix} 13 & 13 & 13 \\ 13 & 13 & 13 \\ 13 & 13 & 13 \end{pmatrix}$
- e. $= \begin{pmatrix} 0 & 2 & 1 \\ 6 & 14 & 6 \\ 2 & 3 & 0 \end{pmatrix}$
- f. $= \begin{pmatrix} 3 & 2 \\ 1 & 0 \end{pmatrix}$
- g. $= \begin{pmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 2 & 3 \\ 0 & 2 & 0 & 3 \\ 4 & 6 & 6 & 9 \end{pmatrix}$
- h. $= \begin{pmatrix} 0 & 0 & 1 \\ 0 & 4 & 6 \\ 4 & 12 & 9 \end{pmatrix}$

Which of the following models or architectures are **one-stage** and which are **two-stage** approaches?

Faster R-CNN	Choose...
YOLO	Choose...
Mask R-CNN	Choose...
Fast R-CNN	Choose...
R-CNN	Choose...
SSD	Choose...

Choose...
✓ Two-stage
One-stage

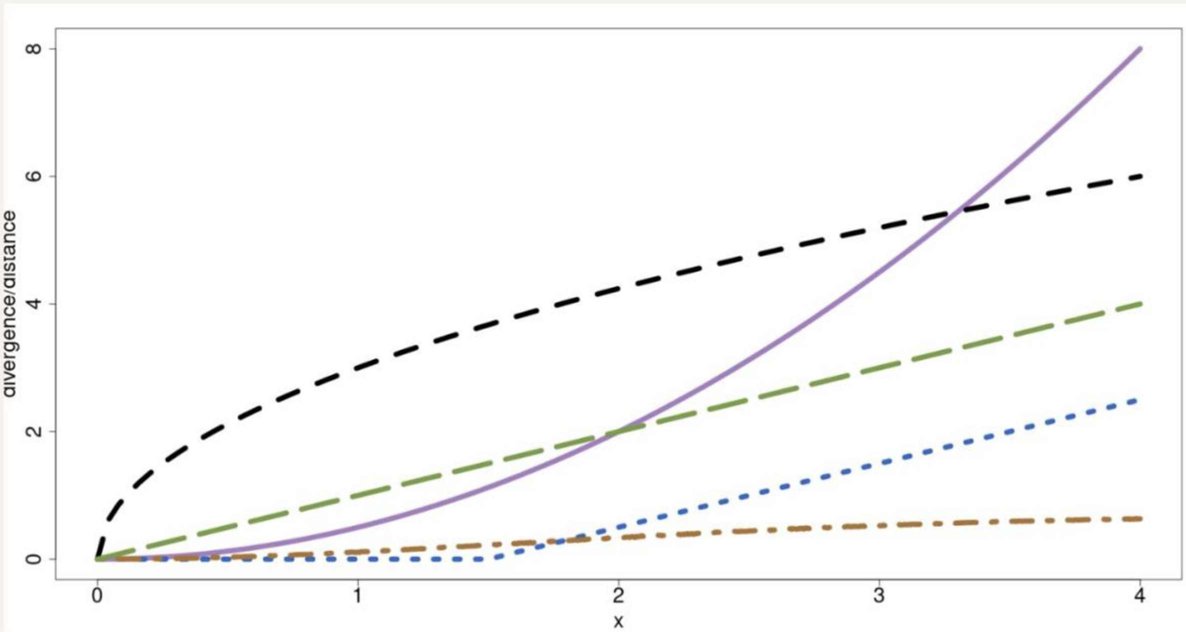
Mark the correct statements concerning the two graphs below



Select one or more:

- [ ] a. $\square_1$ and $\square_2$ can be distinguished by the Weisfeiler-Lehman (color refinement) test
- [ ] b. All message passing neural networks (MPNNs) can distinguish $\square_1$ and $\square_2$
- [ ] c. $\square_1$ and $\square_2$ are isomorphic graphs

Assume you are given two univariate Gaussian distributions. The first one, p_1 has mean zero and unit variance ($\square(0,1)$). The second Gaussian p_2 has also unit variance, but a mean at $x > 0$, hence: $\square(x,1)$. You are interested in the Kullback-Leibler divergence $\mathrm{KL}(p_1||p_2)$, the Jensen-Shannon Divergence $\mathrm{JSD}(p_1||p_2)$, and the Wasserstein-2 distance $\mathrm{WD}(p_1||p_2)$ between these two distributions. You now increase x and plot this value against the divergence or distance measures, which gives three graphs displayed in the figure below. Which divergence belongs to which graph?



Jensen-Shannon divergence: Choose...

Wasserstein distance: Choose...

KL divergence: Choose...

- Choose...
- ✓ Purple (solid) line
- Orange (short-dashed) line
- Black (dashed) line
- Green (long-dashed) line
- Blue (dotted-dashed) line

Consider a Generative Adversarial Network (GAN) which successfully produces images of apples. Which of the following statements are correct?

Select one or more:

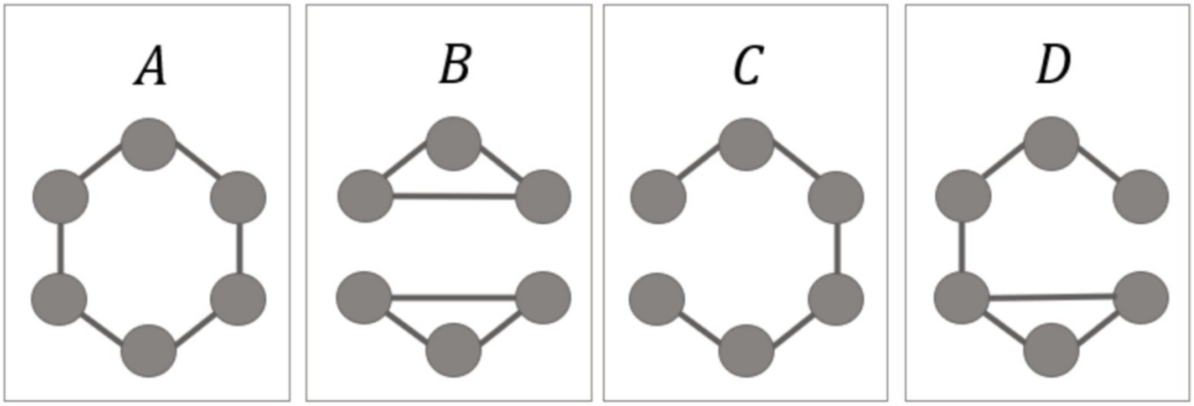
- ☐ a. The discriminator can be used to classify images as apple vs. non-apple.
- ☐ b. The generator aims to learn the distribution of apple images.
- ☐ c. The generator can produce unseen images of apples.
- ☐ d. After training the GAN, the discriminator loss eventually reaches a constant value.

Which of the following statements related to flow-based models are correct?

Select one or more:

- ☐ a. Theoretically, flow-based models can represent any distribution $p_{\mathbf{x}}(\mathbf{x})$ if one starts from a sufficiently complex distribution. A simple base distribution $p_{\mathbf{u}}(\mathbf{u})$ like the uniform distribution on the cube $(0,1)^D$ would not be sufficient.
- ☐ b. Fitting flow based models is often done with the KL-divergence.
- ☐ c. The computation of the Jacobi determinant of a neural network with D inputs and D outputs is in general of $\square(D^3)$.
- ☐ d. For Autoregressive flows the involved transformation g_ϕ is constructed in a way such that its Jacobi determinant always has value 1.

Match the graphs with their corresponding adjacency matrix



$$\begin{bmatrix} 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}$$ Choose...

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$ Choose...

$$\begin{bmatrix} 0 & 1 & 0 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$ Choose...

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$ Choose...

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 \end{bmatrix}$$

Choose...

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

Choose...

(A, B, C, D or none of the above)

Assume you are given an **autoregressive model** $p_w(\boldsymbol{x})$ for sequences (x_1, x_2, x_3). Each element of the sequence can be either A, B, or C.

The model $p_w(\boldsymbol{x})$ is given as follows:

$p_w(x_1) = \frac{1}{3}$ for all $x_1 = A, \ x_2 = B, x_3 = C,$

and

$$p_w(x_2 \mid x_1) = \begin{cases} \frac{1}{2} & \text{if } x_2 = x_1 \\ \frac{1}{4} & \text{if } x_2 \neq x_1 \end{cases},$$

and

$$p_w(x_3 \mid x_2, x_1) = \begin{cases} \frac{1}{2} & \text{if } x_3 = x_2 \\ \frac{1}{4} & \text{if } x_3 \neq x_2 \end{cases}$$

You are given the sequence $\boldsymbol{x} = (A, A, C)$. Calculate $p_w(\boldsymbol{x})$ and enter the result in the box below! **Round the result to three digits after the comma!**

Answer: